

Ming Li

LLM Post-Training, Reasoning & AI Evaluation

PhD Candidate, Computer Science, University of Maryland

Expected Spring 2027

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2026 Apple Scholar in AI/ML | 2× ACL Oral (2025, 2026) | ACL'26 SAC Highlight

I study how to train, understand, and rigorously evaluate LLMs and agentic systems, with a focus on post-training, reasoning behavior, and reliable AI evaluation.

Selected Research

Data-Centric LLM Post-Training

Selecting high-value data and understanding how it shapes LLM adaptation.

- **IFD / Cherry LLM** — NAACL'24. Self-guided data selection achieved competitive instruction tuning with 10% of the data in Alpaca/WizardLM experiments; **470+ citations**.
- **Data Quality through Layer-wise Gradients** — ACL'26. Used layer-wise gradient spectra to explain data-quality metrics across instruction and reasoning post-training. **AC-recommended Oral · Top-15% meta-review**.

Also: Superfiltering (ACL'24), ATLaS (ACL'25 Findings; tuning on 30% of trajectory steps), Reflection-Tuning (ACL'24 Findings). **1K+ GitHub stars** across these post-training codebases.

Reasoning & Model Behavior

Understanding how training and inference reshape the structure and reliability of reasoning.

- **Schoenfeld's Anatomy / ThinkARM** — ACL'26 Oral & SAC Highlight. Functional decomposition of reasoning traces reveals how efficiency interventions alter reasoning structure.
- **Fast vs. Slow Thinking** — ACL'25 Oral. Layer-wise gradient analysis characterizes how direct-answer and explicit-reasoning training differ.

Also: Missing Premise / Overthinking (COLM'25; **16K+ alphaXiv views**); visual reasoning RL analysis (COLM'26).

AI Evaluation & Reliability

Stress-testing models and agents to uncover capability limits, behavioral failures, and unreliable judgments.

- **Moltbook / AI Agent Society** — ACM CAIS'26. Diagnosed agent interaction dynamics, finding persistent individual inertia without stable collective consensus; **Science News interview**.
- **Rhetoric Reward-Hacking** — 2026 preprint. Controlled evaluation of 4,200 manuscript versions from 120 papers reveals how evidence framing and novelty claims shift AI-review judgments.

Also: ColorBench (NeurIPS'25 D&B; co-first); RobustReview / SciCore (2026 manuscript; co-first); ClawEnvKit (research lead; 1,040 agent environments).

Industry & Visiting Research Experience

MBZUAI — Visiting Researcher

Feb–May 2026

Led empirical analysis of social dynamics in AI agent societies (Moltbook; ACM CAIS'26).

Amazon — Applied Scientist Intern

May–Aug 2025

Multi-turn RL / RLAAR (ACL'26): Qwen3-8B retained 75.1% vs. 62.6% of single-turn performance across four LiC tasks.

Microsoft Research — Research Intern

Feb–May 2025

LLM reasoning, interpretability, and collaborative decoding (FoReal-Decoding; NeurIPS'25 workshop).

Adobe — Research Scientist/Engineer Intern

May–Nov 2024

Multimodal grounding / TRIG: 800 annotated QA pairs + ~90K synthetic training instances; ICLR'25 workshop.

Technical Skills: PyTorch · Hugging Face · vLLM · verl · SFT / RLHF / RLVR

Selected Publications

Selected publications and manuscripts · 20+ first/co-first-author papers overall

* Equal contribution

Full list: [Homepage](#) | [Google Scholar](#)

Data-Centric LLM Post-Training

From Quantity to Quality: Boosting LLM Performance with Self-Guided Data Selection for Instruction Tuning.

M. Li, Y. Zhang, Z. Li, et al. — [NAACL 2024](#)

Superfiltering: Weak-to-Strong Data Filtering for Fast Instruction-Tuning.

M. Li, Y. Zhang, S. He, et al. — [ACL 2024](#)

How Instruction and Reasoning Data Shape Post-Training: Data Quality through the Lens of Layer-wise Gradients.

M. Li, Y. Li, Z. Li, T. Zhou — [ACL 2026](#); **AC-recommended Oral · Top-15% meta-review**

Mitigating Lost in Multi-turn Conversation via Curriculum RL with Verifiable Accuracy and Abstention Rewards.

M. Li, P. Chen, Z. Zhang, et al. — [ACL 2026](#)

ATLaS: Agent Tuning via Learning Critical Steps.

Z. Chen*, M. Li*, Y. Huang, et al. — [Findings of ACL 2025](#)

Reasoning & Model Behavior

Schoenfeld’s Anatomy of Mathematical Reasoning by Language Models.

M. Li*, C. Fan*, Y. Cheng*, S. Feizi, T. Zhou — **ACL 2026 Oral & SAC Highlight**

What Happened in LLMs Layers when Trained for Fast vs. Slow Thinking: A Gradient Perspective.

M. Li, Y. Li, T. Zhou — **ACL 2025 Oral**

Missing Premise Exacerbates Overthinking: Are Reasoning Models Losing Critical Thinking Skill?

C. Fan*, M. Li*, L. Sun, T. Zhou — [COLM 2025](#)

What Does RL Improve for Visual Reasoning? A Frankenstein-Style Analysis.

X. Li*, M. Li*, T. Zhou — [COLM 2026](#)

AI Evaluation & Reliability

Does Socialization Emerge in AI Agent Society? A Case Study of Moltbook.

M. Li*, X. Li*, T. Zhou — [ACM CAIS 2026](#)

How Can Rhetoric Reward-Hack AI Reviewers? Dissecting Rhetorical Sensitivity in AI-Based Peer Review.

M. Li*, C. Wang*, X. Li, et al. — 2026 preprint

ColorBench: Can VLMs See and Understand the Colorful World? A Comprehensive Benchmark for Color Perception, Reasoning, and Robustness.

Y. Liang*, M. Li*, C. Fan, et al. — [NeurIPS 2025 Datasets & Benchmarks](#)

BenTo: Benchmark Reduction with In-Context Transferability.

H. Zhao, M. Li, L. Sun, T. Zhou — [ICLR 2025](#)

Research Leadership & Recognition

Recognition. **Best Research Award, e-Assessment Awards 2026** — project lead & first author ([EPM’25](#)) ·

UMD Graduate School Dean’s Fellowship (2023).

Community & Outreach. ACL ARR Area Chair · AI/ML technical content creator, **2.8K followers** on rednote (Xiaohongshu) · [CIKM’25](#) tutorial contributor.

Mentorship. Mentored junior researchers across multiple first-author projects, from problem formulation through publication and independent ownership.

Education

University of Maryland — PhD Candidate, Computer Science

2023–Present

Advisor: Tianyi Zhou | Expected Spring 2027

Texas A&M University — MS, Computer Science

2021–2023

Advisor: Ruihong Huang

Xi’an Jiaotong University — BS, Computer Science

2016–2020